



School of Computation, Information and Technology - Informatics
Technical University of Munich

Bachelor's Thesis in Informatics

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Ein Vergleich von No-Swap-Regret-Algorithmen

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Abstract

External, internal, and swap regret differ in the class of deviations against which a learner is evaluated. In repeated games this distinction also has a game-theoretic interpretation: external regret is linked to coarse correlated equilibrium (CCE), while internal and swap regret are linked to correlated equilibrium (CE). This thesis studies how clearly that distinction appears in finite-horizon experiments.

The implementation covers Hedge, AuerExp3, Exp3-IX, Regret Matching, Stationary Regret Matching, the Blum–Mansour and Ito reductions, Tsallis-INF as Ito’s bandit sublearner, and LCE-IX. Full-information and bandit experiments are run in RPS and RPSLS self-play, an adaptive historical-frequency environment, and an independent lazy random walk. All algorithms are evaluated with the same external, internal, and swap-regret metrics; self-play runs are additionally evaluated by the L_1 distance of the empirical joint-action distribution to the CE and CCE sets.

The current eight-replicate results show a strong environment dependence. The historical-frequency adversary separates the algorithms clearly, with substantially larger internal and swap regret for the external-regret baselines and for ordinary Regret Matching. Under the independent random walk the differences are much smaller, particularly with bandit feedback. In self-play, the equilibrium-distance plots provide a direct example of the distinction between CCE and CE: full-information Hedge can be close to CCE while remaining farther from CE, whereas the internal- and swap-regret procedures generally approach CE more closely in the games studied. The results are preliminary finite-horizon evidence rather than an empirical estimate of asymptotic regret rates.

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1. Introduction

1.1. Motivation

Regret minimization is one of the standard ways to study repeated decision making. In a finite game, the regret notion used by the players also determines which equilibrium set is reached by empirical play: external regret is associated with coarse correlated equilibrium (CCE), while internal and swap regret lead to correlated equilibrium (CE) [1–4]. The distinction is simple in theory, but it does not immediately tell us what the algorithms look like at a finite horizon.

This thesis is mainly experimental. It compares several well-known external-, internal-, and swap-regret algorithms under the same evaluation code and asks when the stronger regret guarantees are visible in practice. The comparison includes both full-information and bandit feedback. In addition to one-player time-varying environments, the algorithms are run in self-play in Rock–Paper–Scissors (RPS) and Rock–Paper–Scissors–Lizard–Spock (RPSLS), where regret can also be related to CE and CCE convergence. Recent work on no-swap-regret self-play gives further motivation for studying these dynamics beyond worst-case regret bounds [5].

1.2. Questions and Scope

The experiments are organized around two questions:

1. How different are external-, internal-, and swap-regret algorithms at a finite horizon, and how much does the answer depend on the feedback model and the environment?
2. In self-play, do the empirical distributions show the expected distinction between convergence to CCE and convergence to CE?

The thesis does not propose a new regret bound. The theoretical chapters summarize the results needed to interpret the experiments and document the exact algorithmic instantiations used in the code. The main contribution is the empirical comparison itself: the same regret evaluator is applied to all learners, and repeated-game runs are additionally evaluated by their distance to the full CE and CCE sets.

1.3. Thesis Structure

Chapter 2 introduces the regret notions and the equilibrium concepts used later. Chapter 3 describes the implemented algorithms and their parameter choices. Chapter 4 gives the

experimental protocol and environments. The results are presented in Chapter 5, followed by a short discussion of the main observations and limitations in Chapter 6. Chapter 7 concludes the thesis.

2. Theoretical Background

2.1. Online Learning and Feedback

We consider a finite action set

$$[K] = \{1, \dots, K\}.$$

At round t , the learner chooses a probability distribution

$$p_t \in \Delta_K = \left\{ p \in \mathbb{R}_{\geq 0}^K : \sum_{i=1}^K p(i) = 1 \right\},$$

samples an action $I_t \sim p_t$, and receives reward $r_t(I_t)$, where

$$r_t = (r_t(1), \dots, r_t(K)) \in [0, 1]^K.$$

The learner chooses p_t before observing the feedback of round t . This is the **standard finite-action online-learning model** [6, 7].

The reward sequence need not be independent over time. In particular, r_t may depend on the interaction history before round t . This allows adaptive environments such as the historical-frequency adversary used later in the experiments. We assume, however, that the current reward vector does not depend on the learner's fresh random draw I_t ; the environment is therefore non-anticipating [6, 7].

Randomization. **Randomization is important in adversarial online learning.** A deterministic learner can become predictable from the history, whereas a randomized learner commits only to the distribution p_t before its current action is realized. Conditional on p_t and r_t , the learner's expected reward in round t is

$$\mathbb{E}[r_t(I_t) \mid p_t, r_t] = \langle p_t, r_t \rangle.$$

This quantity is used below in the strategy-weighted regret measures.

Full-information feedback. After the action is chosen, the learner observes the complete reward vector r_t . It can therefore evaluate all actions when performing its update.

Bandit feedback. The learner observes only the realized reward

$$r_t(I_t),$$

while the rewards of unplayed actions remain hidden. Bandit algorithms must therefore estimate missing rewards or losses from the sampled action and its sampling probability. The bandit algorithms considered in this thesis use standard importance-weighted or implicit-exploration constructions [8–10].

The feedback restriction applies to the learner, not to offline evaluation. In the experiments, the simulator retains the complete payoff vector in order to compute counterfactual regret, but a bandit learner receives only the reward of its sampled action.

Throughout the thesis we use rewards rather than losses. Much of the online-learning literature instead uses

$$\ell_t(i) = 1 - r_t(i).$$

For rewards in $[0, 1]$, the two conventions are equivalent, since

$$r_t(j) - r_t(i) = \ell_t(i) - \ell_t(j).$$

The reward convention is convenient for the repeated games and adversarial environments used in the experiments.

2.2. Regret Notions

Regret compares the learner’s reward with the reward that would have been obtained after applying a class of alternative decisions to the same sequence of reward vectors [3, 4, 6]. Different classes of deviations give rise to external, internal, and swap regret.

For randomized learners, this thesis reports both realized and strategy-weighted regret. Realized regret uses the sampled action I_t . Strategy-weighted regret instead averages the one-round deviation gain with respect to the current mixed strategy p_t . The implementation and figures label the latter quantity *expected regret*. We use the term *strategy-weighted* in the theoretical discussion to distinguish it from pseudo-regret and from expectations taken over an entire randomized trajectory.

External regret. External regret compares the learner with the best fixed action in hindsight:

$$R_T^{\text{ext}} = \max_{j \in [K]} \sum_{t=1}^T (r_t(j) - r_t(I_t)).$$

The corresponding strategy-weighted quantity is

$$\bar{R}_T^{\text{ext}} = \max_{j \in [K]} \sum_{t=1}^T (r_t(j) - \langle p_t, r_t \rangle).$$

An algorithm is no-external-regret if its average external regret vanishes under the relevant deterministic, expected, or high-probability guarantee. Standard exponential-weights methods achieve $O(\sqrt{T \log K})$ external regret in the full-information setting [6].

Internal regret. For two actions $i, j \in [K]$, consider the replacement that changes every occurrence of action i to action j and leaves all other actions unchanged. Its realized replacement regret is

$$R_T^{i \rightarrow j} = \sum_{t=1}^T \mathbf{1}\{I_t = i\} (r_t(j) - r_t(i)),$$

and the strategy-weighted analogue is

$$\bar{R}_T^{i \rightarrow j} = \sum_{t=1}^T p_t(i) (r_t(j) - r_t(i)).$$

Internal regret is the largest gain obtainable by one such replacement,

$$R_T^{\text{int}} = \max_{i, j \in [K]} R_T^{i \rightarrow j}.$$

This is the regret notion underlying regret matching and its connection to correlated equilibrium [1–3].

Swap regret. Swap regret allows a different replacement for every source action. Let $\phi : [K] \rightarrow [K]$ be a mapping from played actions to replacement actions. Then

$$R_T^{\text{swap}} = \max_{\phi: [K] \rightarrow [K]} \sum_{t=1}^T (r_t(\phi(I_t)) - r_t(I_t)).$$

Because the replacement $\phi(i)$ can be optimized independently for each source action,

$$R_T^{\text{swap}} = \sum_{i=1}^K \max_{j \in [K]} R_T^{i \rightarrow j}.$$

The same identity holds for the strategy-weighted version.

External and internal deviations are special cases of swap deviations, so

$$R_T^{\text{ext}} \leq R_T^{\text{swap}}, \quad R_T^{\text{int}} \leq R_T^{\text{swap}}.$$

Moreover,

$$R_T^{\text{swap}} \leq K R_T^{\text{int}},$$

so for fixed K , vanishing internal regret and vanishing swap regret are equivalent up to a factor depending on the number of actions [4].

Regret notion	Deviation	Basic alternatives
External	Replace all actions by one fixed action	K
Internal	Replace one action i by one action j	$K(K - 1)$
Swap	Choose a replacement for every source action	K^K

2.3. Repeated Games and Equilibrium Concepts

Consider a finite normal-form game with players $i \in N$, action sets A_i , and payoffs $u_i(a) \in [0, 1]$. After T rounds, the empirical distribution of joint actions is

$$\hat{\mu}_T(a) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{a_t = a\}.$$

This distribution records how frequently each joint action profile occurred during learning.

Coarse correlated equilibrium. A distribution μ is a coarse correlated equilibrium (CCE) if no player can benefit by committing in advance to one fixed alternative action [2, 11]. For every player i and action a'_i ,

$$\sum_a \mu(a) (u_i(a'_i, a_{-i}) - u_i(a)) \leq 0.$$

The deviation is chosen without conditioning on the action assigned to the player, which is why CCE is naturally associated with external regret.

Correlated equilibrium. A distribution μ is a correlated equilibrium (CE) if a player cannot improve by replacing one recommended action with another [12]. For all $a_i, a'_i \in A_i$,

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) (u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i})) \leq 0.$$

The deviation can depend on the recommended action, which corresponds directly to the action-dependent replacements used in internal regret. Every CE is also a CCE,

$$\text{CE} \subseteq \text{CCE},$$

but the converse need not hold. This distinction is relevant in RPS and RPSLS self-play, where the empirical distribution may become close to the larger CCE set without being equally close to CE [5].

2.4. Regret and Equilibrium Convergence

The link between regret minimization and equilibrium learning follows by comparing the regret inequalities with the incentive constraints above [1–3].

If every player has external regret at most $\varepsilon_T T$, then for every fixed deviation a'_i ,

$$\sum_a \hat{\mu}_T(a) (u_i(a'_i, a_{-i}) - u_i(a)) \leq \varepsilon_T.$$

Thus $\hat{\mu}_T$ is an ε_T -CCE. Vanishing average external regret therefore implies that empirical play approaches the CCE set.

Replacement regret gives the analogous statement for CE. If every replacement regret of every player is at most $\varepsilon_T T$, then

$$\sum_{a_{-i}} \hat{\mu}_T(a_i, a_{-i}) (u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i})) \leq \varepsilon_T$$

for all a_i, a'_i . Hence vanishing internal regret implies convergence of empirical play toward CE. Since vanishing swap regret implies vanishing internal regret, the same conclusion applies to no-swap-regret algorithms [4].

These statements concern the regret notion actually controlled by the algorithm. Some bandit results in the literature are stated for pseudo-regret, some in expectation, and others with high probability. Chapter 3 records the form of the guarantee used for each algorithm [4, 13, 14].

The distinction motivates the self-play experiments:

$$\text{external regret} \implies \text{CCE},$$

whereas

$$\text{internal or swap regret} \implies \text{CE}.$$

The experiments measure both regret and distance of empirical play to the CE and CCE sets; the latter metric is defined in Chapter 4.

3. Regret-Minimizing Algorithms

The experiments use standard algorithms from the literature rather than new variants. This chapter records the update rules and, where relevant, the parameter choices used in the implementation.

3.1. External-Regret Algorithms

Hedge. Hedge is the full-information exponential-weights baseline [6, 15]. With cumulative reward $G_{t-1}(i) = \sum_{s < t} r_s(i)$,

$$p_t(i) = \frac{\exp(\eta G_{t-1}(i))}{\sum_j \exp(\eta G_{t-1}(j))}.$$

For a known horizon, the implementation uses

$$\eta = \sqrt{\frac{8 \log K}{T}},$$

which gives the usual $O(\sqrt{T \log K})$ external-regret order for rewards in $[0, 1]$. Standalone Hedge and the Hedge learners inside Blum–Mansour use this fixed-horizon choice. The full-information Ito reduction instead uses an anytime schedule $\eta_s = \sqrt{8 \log K / s}$ on the local time s of each sublearner.

AuerExp3. The bandit baseline follows Exp3 [8]. If q_t is the exponential-weights distribution, the played distribution is

$$p_t(i) = (1 - \gamma)q_t(i) + \frac{\gamma}{K},$$

and the observed gain is estimated by

$$\hat{r}_t(i) = \frac{r_t(I_t) \mathbf{1}\{I_t = i\}}{p_t(i)}.$$

The experiments use the parameter choice employed in the Blum–Mansour bandit reduction [4],

$$\gamma = \min \left\{ 1, \sqrt{\frac{K \log K}{T}} \right\}, \quad \eta = \frac{\gamma}{K}.$$

The same choice is used for the standalone baseline, called *AuerExp3* in the figures. It is therefore aligned with the Blum–Mansour implementation rather than with the constant-optimized standalone corollary in Auer et al. [8].

Exp3-IX. Exp3-IX uses implicit rather than explicit exploration [9]. With loss $\ell_t(i) = 1 - r_t(i)$,

$$\widehat{\ell}_t(i) = \frac{\ell_t(I_t) \mathbf{1}\{I_t = i\}}{p_t(i) + \gamma}.$$

The implementation uses

$$\eta = \sqrt{\frac{2 \log K}{KT}}, \quad \gamma = \frac{\eta}{2},$$

matching the parameterization in the high-probability analysis of Neu [9]. No additional probability floor is added to the exponential-weights distribution.

Tsallis-INF. The bandit Ito reduction uses the anytime 1/2-Tsallis-INF algorithm of Zimmert and Seldin [10]. It employs the standard importance-weighted loss estimator and the local learning rate

$$\eta_s = \frac{1}{\sqrt{s}}.$$

The implementation computes the Tsallis distribution from the corresponding KKT equation. After shifting the cumulative estimated losses, the normalizer is the unique $z \in [1, \sqrt{K}]$ satisfying

$$\sum_{i=1}^K (a_i + z)^{-2} = 1,$$

with probabilities $p_s(i) = (a_i + z)^{-2}$. A safeguarded Newton–bisection routine is used only to solve this one-dimensional equation numerically.

3.2. Regret-Matching Procedures

Hart and Mas-Colell study two closely related procedures built from cumulative pairwise replacement gains [3]. Let

$$C_t(i, j) = \sum_{s=1}^t \mathbf{1}\{I_s = i\} (r_s(j) - r_s(i)).$$

For $i \neq j$, define

$$Q_t(i, j) = \frac{[C_t(i, j)]_+}{Kt},$$

and assign the remaining probability mass to the diagonal,

$$Q_t(i, i) = 1 - \sum_{j \neq i} Q_t(i, j).$$

The two procedures use the same matrix Q_t , but they turn it into the next strategy in different ways. This difference is important both theoretically and in the experiments.

Regret Matching (RM). The original regret-matching procedure uses the row associated with the action just played. If $I_t = i$, then

$$p_{t+1} = Q_t(i, \cdot).$$

This is the procedure in Section 2 of Hart and Mas-Colell [3]. Its main guarantee is a *joint self-play* statement: if every player in the repeated game follows regret matching, then the empirical distribution of play converges almost surely to the set of correlated equilibria. The theorem is not an individual online-learning guarantee stating that one RM player has vanishing internal regret against an arbitrary adaptive opponent. Consequently, the one-player adversarial experiments in this thesis should not be interpreted as testing a guarantee that the original RM procedure does not claim.

This distinction also matters for external regret. RM is included because of its role in correlated-equilibrium learning, not because it is a standalone external-regret algorithm. In the self-play experiments, the relevant theoretical prediction is convergence of the joint empirical distribution to CE when all players use RM. The regret curves are reported as additional finite-horizon diagnostics.

Stationary Regret Matching (SRM). Section 3 of Hart and Mas-Colell [3] considers a different strategy selection rule. Instead of following one row of Q_t , the player chooses a stationary distribution

$$p_{t+1} = p_{t+1} Q_t.$$

We refer to this procedure as *Stationary Regret Matching* (SRM). Theorem A of Hart and Mas-Colell [3] gives an individual-player result: the pairwise replacement regrets vanish without requiring the opponents to use the same procedure. This is the guarantee relevant to the one-player adversarial experiments.

For a fixed finite action set, the relation

$$R_T^{\text{swap}} \leq K R_T^{\text{int}}$$

from Chapter 2 implies that vanishing average internal regret also yields vanishing average swap regret. Thus SRM is not introduced in the original paper as a “swap-regret algorithm”, but its internal-regret guarantee is sufficient for no-swap-regret when K is fixed. The experiments report all three regret notions in order to make this consequence visible.

3.3. Swap-Regret Algorithms

The remaining algorithms maintain K external-regret sublearners. If sublearner i outputs $q_t^{(i)}$, the rows form a stochastic matrix Q_t . The master distribution is chosen to satisfy

$$p_t = p_t Q_t.$$

This stationary reduction is the common structure behind Blum–Mansour, Ito, and LCE-IX [4, 13, 14].

Blum–Mansour. Blum–Mansour (BM) updates all K sublearners every round [4]. Under full information, sublearner i receives the reward vector

$$p_t(i)r_t.$$

The present implementation uses standard fixed-horizon Hedge as the base learner. This is a valid instance of the generic reduction, but it is not the first-order experts instantiation used for the sharpest full-information bound in Blum and Mansour [4]; with standard Hedge, the direct composition gives the more conservative order $O(K\sqrt{T\log K})$.

Under bandit feedback, the sublearners are AuerExp3 instances. If the realized master action is I_t , sublearner i receives

$$\frac{p_t(i)q_t^{(i)}(I_t)}{p_t(I_t)}r_t(I_t).$$

Together with the inner Exp3 importance weighting, this matches the partial-information construction of Blum and Mansour [4], which gives swap pseudo-regret of order

$$O\left(\sqrt{TK^3\log K}\right).$$

Ito reduction. Ito’s reduction changes the update schedule of the sublearners [13]. It samples

$$J_t \sim p_t, \quad I_t \sim q_t^{(J_t)},$$

and updates only sublearner J_t . If the base learner has a concave external-regret bound $r(\cdot)$, Ito proves

$$\mathbb{E}[R_T^{\text{swap}}] \leq Kr(T/K),$$

where the expectation is over the algorithmic randomization [13]. With anytime Hedge this yields $O(\sqrt{TK\log K})$ in the full-information implementation. With Tsallis-INF, whose adversarial bandit regret is $O(\sqrt{Ks})$ [10], the resulting bandit order is $O(K\sqrt{T})$.

LCE-IX. LCE-IX is a bandit swap-regret algorithm based on the same stationary architecture but with implicit exploration [14]. All K sublearners are updated every round. For the realized action I_t , row i receives the loss quantity

$$\gamma_t^{(i)} = \frac{p_t(i)q_t^{(i)}(I_t)}{p_t(I_t)}(1 - r_t(I_t)),$$

which is divided by $q_t^{(i)}(I_t) + \gamma_t$ in the implicit-exploration estimator. The implementation uses

$$\eta_t = \sqrt{\frac{\log K}{t}}, \quad \gamma_t = \frac{\eta_t}{2},$$

as in Huang and Pan [14]. The cited analysis gives a near-optimal high-probability swap-regret bound of order $K\sqrt{T\log(K/\delta)}$, and the corresponding expected order is $O(K\sqrt{T\log K})$.

3.4. Algorithms Used in the Experiments

Table 3.1.: Algorithms used in the experiments. The column “Role / guarantee” summarizes the reason for including each method; it does not imply that all guarantees are of the same probabilistic type.

Algorithm	Feedback	Role / guarantee	Main ingredient
Hedge	Full	External regret	Exponential weights
AuerExp3	Bandit	External regret	Explicit exploration
Exp3-IX	Bandit	External regret	Implicit exploration
RM	Full	CE in joint self-play	Current regret row
SRM	Full	Individual internal regret	Stationary regret matrix
BM	Full / bandit	Swap-regret reduction	Update all sublearners
Ito	Full / bandit	Swap-regret reduction	Update selected sublearner
LCE-IX	Bandit	Swap regret	Stationary reduction + IX

The numerical implementation follows these equations. Stationary distributions are obtained by solving $pQ = p$ together with $\sum_i p_i = 1$, and the Tsallis-INF normalizer is obtained from the KKT equation above. These numerical routines affect how the equations are solved, not the definitions of the algorithms.

4. Experimental Methodology

4.1. Experimental Setup

All experiments use rewards in $[0, 1]$ and a final horizon of

$$T = 10^6.$$

Each configuration is run for **eight replicates**. The base seed is 42, with replicate-specific learner and environment seeds derived deterministically from it. The figures show **replicate means and pointwise two-sided 95% Student- t confidence intervals**.

Full-information experiments compare Hedge, BM, Ito, Regret Matching (RM), and Stationary Regret Matching (SRM). Bandit experiments compare AuerExp3, Exp3-IX, BM, Ito, and LCE-IX. In RPS and RPSLS, both players use the same algorithm. The repeated-game regret plots in Chapter 5 use player 0, whereas the equilibrium-distance calculation uses the complete joint-action history.

At each round the learner samples from its current strategy, the environment produces the reward, and the learner receives either the complete reward vector or the sampled reward according to the feedback model. Regret is evaluated separately from the learner update. In particular, bandit learners never receive counterfactual rewards even though the simulator retains them for offline regret evaluation.

Table 4.1.: Experiment groups used in this draft.

Environment	Setting	Actions	Feedback
RPS	Two-player self-play	3 per player	Full / bandit
RPSLS	Two-player self-play	5 per player	Full / bandit
Historical-frequency adversary	One player	9	Full / bandit
Independent lazy random walk	One player	9	Full / bandit

4.2. Environments

Rock–Paper–Scissors and RPSLS. RPS and RPSLS are symmetric constant-sum games. The implementation stores normalized payoffs with win = 1, tie = 0.5, and loss = 0. In RPSLS the action order is Rock, Paper, Scissors, Lizard, Spock; each action beats two and loses to two. The five-action game follows the cyclic example used in the self-play study of Paes Leme et al. [5]. RPS gives the smallest cyclic benchmark, while RPSLS provides a similar setting with a larger action space.

Historical-frequency adversary. The historical-frequency adversary is an adaptive one-player environment. Before the current action is added to the history, the $\lceil K/2 \rceil$ most frequently played actions receive reward 0, and the remaining actions receive reward 1. For $K = 9$, five actions are punished each round. Ties are resolved by a deterministic rotating rule. The construction depends on past play but not on the learner’s current random draw. It is used as a stress test for action-dependent regret rather than as a model of a particular application.

Independent lazy random walk. The second one-player environment assigns an independent lazy random walk to every action. Rewards lie on the grid

$$\{0, 0.1, \dots, 1\}$$

and all actions start at 0.5. At an interior point, the process moves down, stays, or moves up with probability $1/3$ each. At a boundary, it stays or moves one step inward with probability $1/2$. The reward process is generated independently of the learner’s actions. Within a replicate, the same environment seed is used across algorithms so that they face the same reward trajectory.

4.3. Evaluation Metrics

Regret. The experiment code maintains one **cumulative replacement-gain matrix**. For strategy-weighted regret,

$$G_T(i, j) = \sum_{t=1}^T p_t(i) (r_t(j) - r_t(i)),$$

while realized regret replaces $p_t(i)$ by $\mathbf{1}\{I_t = i\}$. External, internal, and swap regret are then read from the same matrix:

$$R_T^{\text{ext}} = \max_j \sum_i G_T(i, j), \quad R_T^{\text{int}} = \max_{i,j} G_T(i, j),$$

$$R_T^{\text{swap}} = \sum_i \max_j G_T(i, j).$$

These are the replacement-regret quantities introduced in Chapter 2 [4].

The reported full-information figures use the strategy-weighted version, labeled *expected* in the plots. Bandit figures use realized regret. Two normalizations are retained in the result archive: average regret R_t/t , which corresponds directly to the no-regret criterion, and $R_t/\sqrt{t}$, which is used as a finite-horizon diagnostic. For fixed-horizon algorithms, an intermediate point on a $T = 10^6$ run is not the same as an independently tuned run whose final horizon equals that intermediate value.

Equilibrium distance. For RPS and RPSLS, the complete action history is used to form the empirical distribution $\hat{\mu}_t$. We measure

$$d_{\text{CE}}(\hat{\mu}_t) = \min_{\mu \in \text{CE}} \|\hat{\mu}_t - \mu\|_1,$$

and

$$d_{\text{CCE}}(\hat{\mu}_t) = \min_{\mu \in \text{CCE}} \|\hat{\mu}_t - \mu\|_1.$$

Because $\text{CE} \subseteq \text{CCE}$, the CCE distance cannot exceed the CE distance. The minimizations are formulated as linear programs over the full joint-action distribution and solved through SciPy’s HiGHS backend [16, 17]. At most 160 approximately log-spaced horizons are evaluated per trajectory; each selected point uses the exact empirical distribution at that horizon.

The long runs are recorded at a limited number of deterministic checkpoints, while learner and regret states are updated every round. Repeated-game action histories are stored losslessly so that the empirical joint-action distribution can be reconstructed exactly.

5. Experimental Results

This chapter presents the complete set of figures from the current experiment archive. No result type is omitted: for every experiment we show external, internal, and swap regret, both as average regret R_t/t and normalized by $\sqrt{t}$. For RPS and RPSLS self-play we additionally show the CE/CCE distance trajectory of every algorithm. The purpose of the chapter is first to report what is visible in the data and then to relate the main patterns to the guarantees summarized in Chapters 2 and 3.

All curves are means over eight replicates with base seed 42. Shaded regions denote point-wise 95% Student- t confidence intervals. Full-information figures use the strategy-weighted regret reported as “expected” in the implementation, whereas bandit figures use realized regret. The normalized curves $R_t/\sqrt{t}$ are trajectories within runs whose final horizon is $T = 10^6$; they are not independent estimates of an asymptotic exponent.

5.1. Full-Information Experiments

Historical-frequency adversary. Figures 5.1 and 5.2 contain all six regret views for the nine-action historical-frequency adversary. The environment separates the algorithms clearly. Hedge has the most favorable external-regret profile among the simple baselines, but its internal and swap regret are much larger. RM also accumulates substantial regret in this one-player adaptive setting. This is not in conflict with Hart and Mas-Colell [3]: the original RM procedure is accompanied by a joint self-play CE-convergence theorem, not by an individual adversarial no-internal-regret guarantee. SRM, whose stationary version does have an individual pairwise-regret guarantee, is much lower on the internal and swap measures. BM and Ito also remain low in swap regret, with Ito generally reaching its low-regret regime earlier than BM in this experiment.

Independent lazy random walk. The same six views for the independent lazy random walk are shown in Figures 5.3 and 5.4. The differences are less dramatic than in the historical-frequency adversary, although the qualitative roles remain recognizable: Hedge is strongest on external regret, while SRM, BM, and Ito behave better on the stronger deviation classes. RM again has noticeably larger finite-horizon internal and swap regret than SRM. The random-walk rewards are independent of the learner’s actions, so the environment does not react strategically to predictable behavior. This makes it a useful contrast to the historical-frequency experiment.

5. Experimental Results

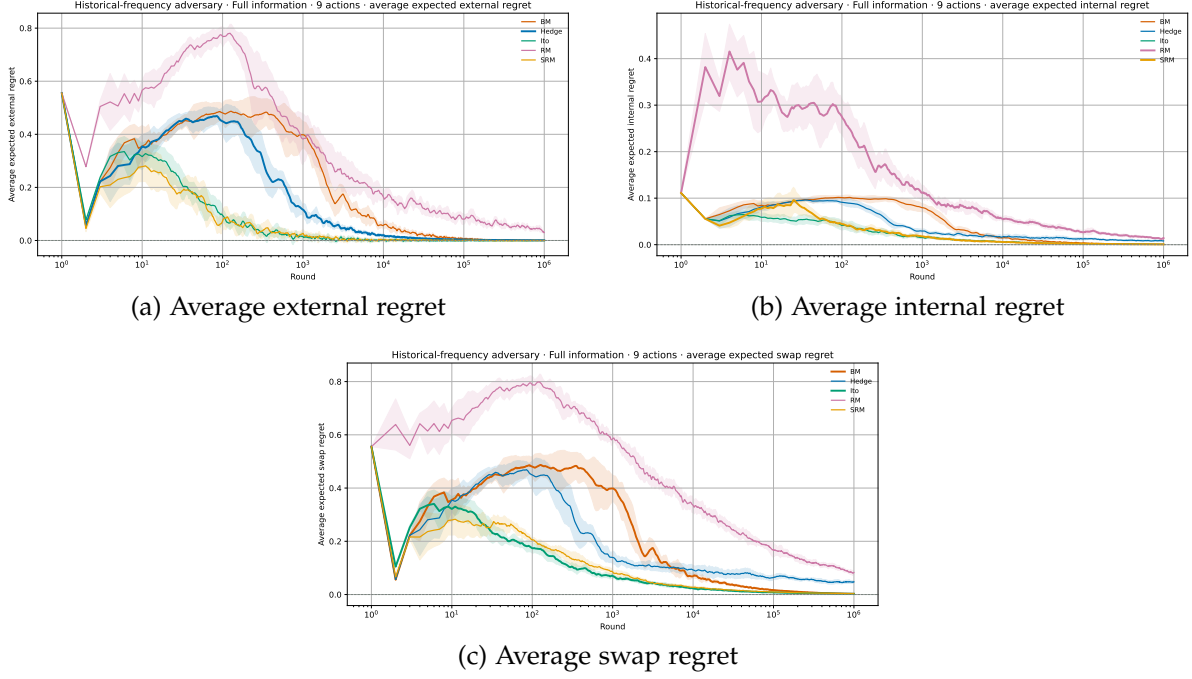


Figure 5.1.: Historical-frequency adversary, full information, $K = 9$, $T = 10^6$, eight replicates: complete average strategy-weighted regret results.

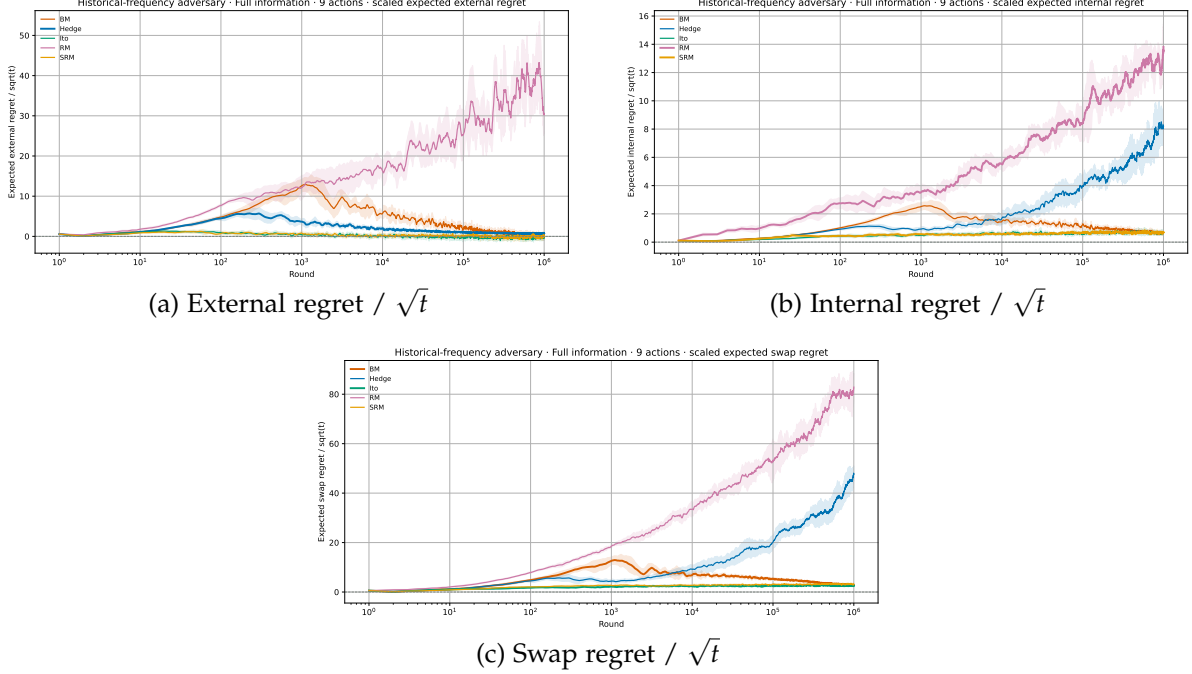


Figure 5.2.: Historical-frequency adversary, full information: normalized strategy-weighted regret. The large RM–SRM separation is visible in both internal and swap regret.

5. Experimental Results

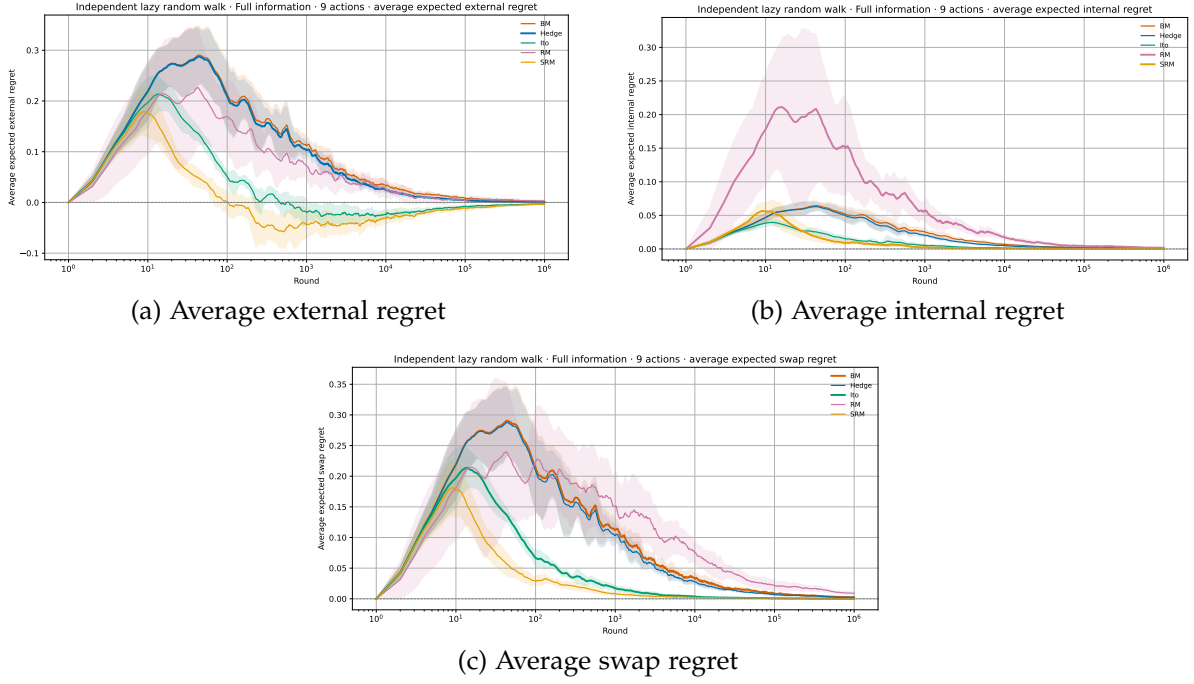


Figure 5.3.: Independent lazy random walk, full information, $K = 9$, $T = 10^6$, eight replicates: complete average strategy-weighted regret results.

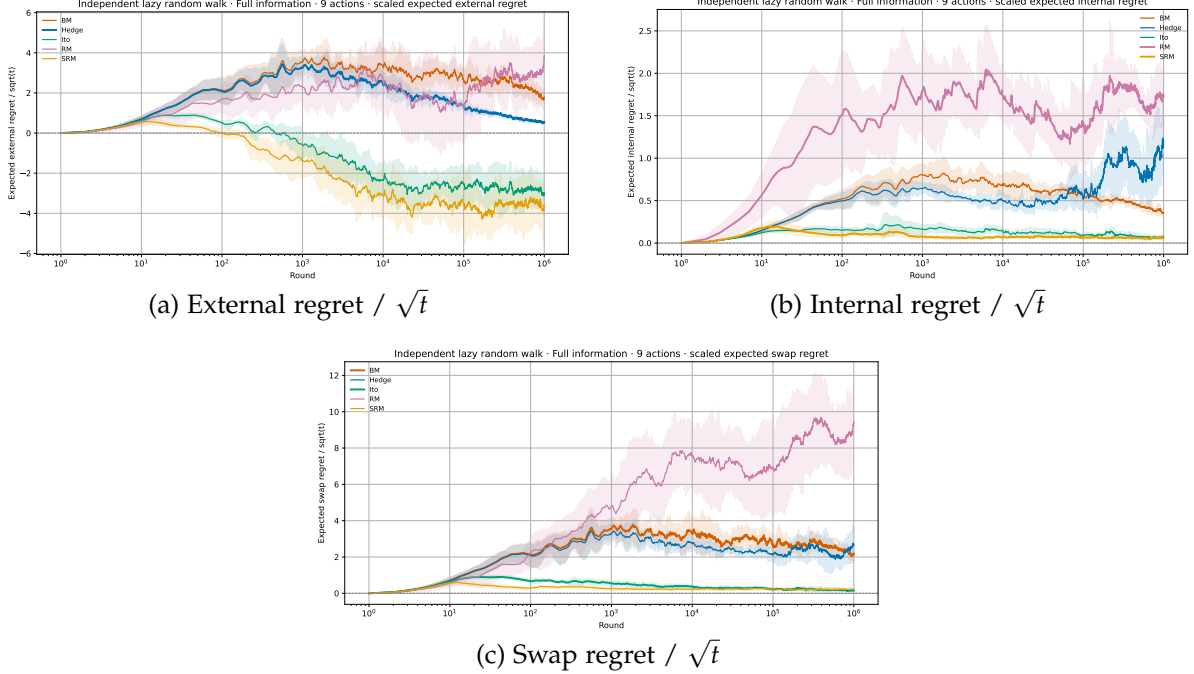


Figure 5.4.: Independent lazy random walk, full information: complete normalized strategy-weighted regret results.

RPS self-play. Figures 5.5–5.7 report every RPS full-information result for both players. The regret curves show the same distinction as the one-player experiments. Hedge keeps external regret comparatively small while its internal and swap regret are substantially larger. BM and Ito, although built from Hedge-type external-regret sublearners, control swap regret much more effectively after the stationary reduction. SRM is again among the fastest-converging methods in the present runs.

RM is particularly informative here. Its finite-horizon external, internal, and swap regret can remain sizable, yet the empirical distribution of RM self-play moves toward the CE set. This is the behavior predicted by the Main Theorem of Hart and Mas-Colell [3], which assumes that all players use the original RM procedure. The experiment therefore illustrates the distinction between a joint CE-convergence result and an individual adversarial regret guarantee.

The equilibrium panels also separate Hedge from the stronger procedures. Hedge becomes very close to CCE while its CE distance remains visibly larger. BM, Ito, RM, and SRM approach CE more directly, with SRM particularly fast in this game.

RPSLS self-play. The five-action RPSLS game gives a similar but generally slower version of the RPS behavior (Figures 5.8–5.10). Hedge again has a clear external-regret advantage over its own internal and swap performance, and the equilibrium-distance plot shows a marked CCE–CE gap at long horizons. BM and Ito turn the same type of external-regret base learner into procedures with much smaller swap regret and CE-oriented self-play. SRM remains very fast on both the regret and equilibrium measures.

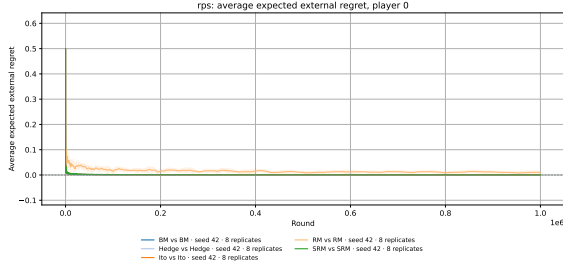
RM approaches CE here as well, but more slowly than in RPS. Its behavior is again consistent with the joint self-play result of Hart and Mas-Colell [3]; the experiment does not require RM to satisfy an individual adversarial internal-regret theorem. Across the current full-information runs, Ito usually reaches low swap regret faster than BM, although the figures are finite-horizon comparisons rather than a general ordering theorem.

5.2. Bandit Experiments

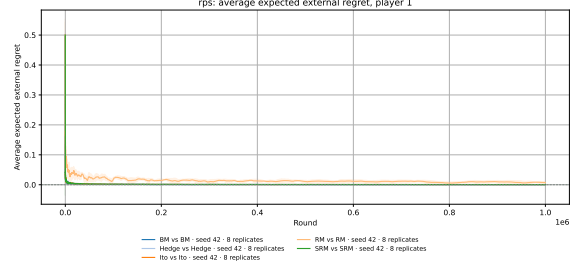
Historical-frequency adversary. The bandit historical-frequency experiment gives the cleanest separation between the standalone external-regret baselines and the dedicated swap-regret methods (Figures 5.11 and 5.12). AuerExp3 and Exp3-IX have much larger realized swap regret over the tested horizon. This does not prove that either algorithm has linear swap regret; rather, neither algorithm is designed with a swap-regret guarantee, and this environment makes the missing guarantee visible at finite horizon. BM, Ito, and LCE-IX remain much lower. BM shows a pronounced transient before decreasing, while LCE-IX has the lowest observed mean near the final horizon.

Independent lazy random walk. In the lazy random walk, the five bandit methods are much harder to separate (Figures 5.13 and 5.14). Even AuerExp3 and Exp3-IX show low observed swap regret relative to the historical-frequency experiment, and the dedicated swap-

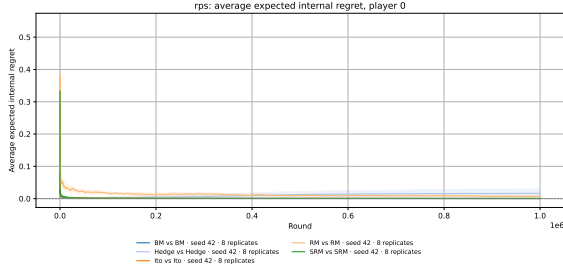
5. Experimental Results



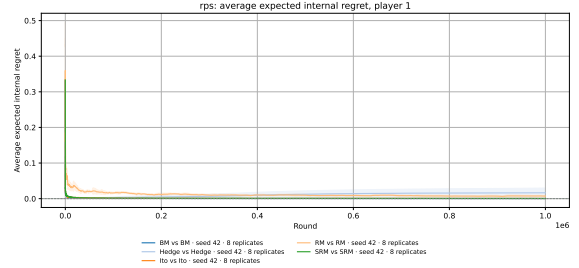
(a) Player 0: average external



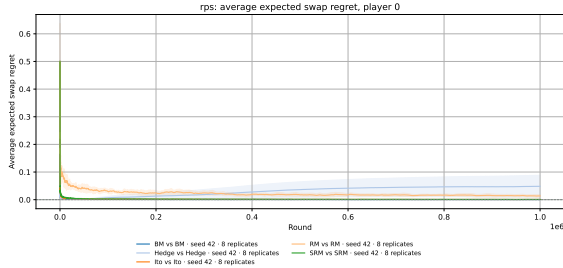
(b) Player 1: average external



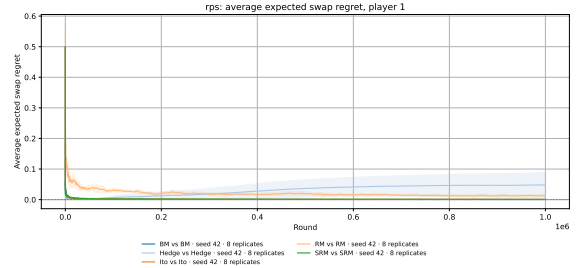
(c) Player 0: average internal



(d) Player 1: average internal



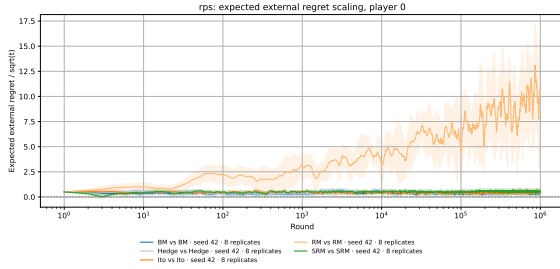
(e) Player 0: average swap



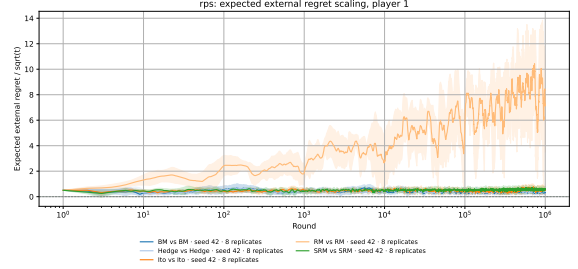
(f) Player 1: average swap

Figure 5.5.: RPS self-play, full information, $T = 10^6$, eight replicates: all average strategy-weighted regret results. The two columns correspond to the two players; rows show external, internal, and swap regret.

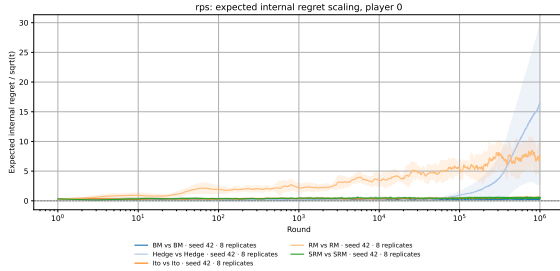
5. Experimental Results



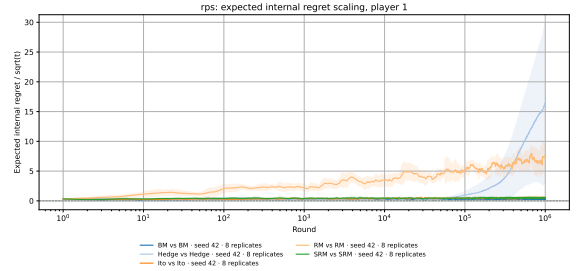
(a) Player 0: external / $\sqrt{t}$



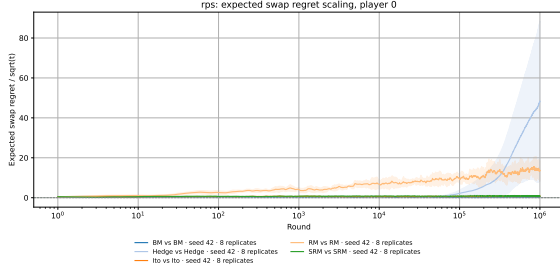
(b) Player 1: external / $\sqrt{t}$



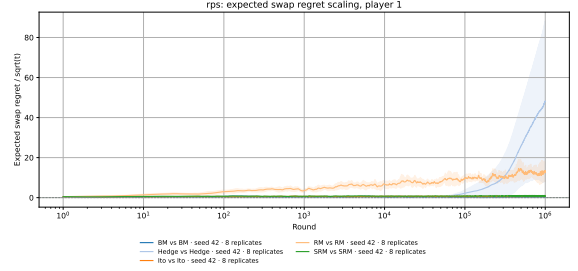
(c) Player 0: internal / $\sqrt{t}$



(d) Player 1: internal / $\sqrt{t}$



(e) Player 0: swap / $\sqrt{t}$



(f) Player 1: swap / $\sqrt{t}$

Figure 5.6.: RPS self-play, full information: all normalized strategy-weighted regret results.

5. Experimental Results

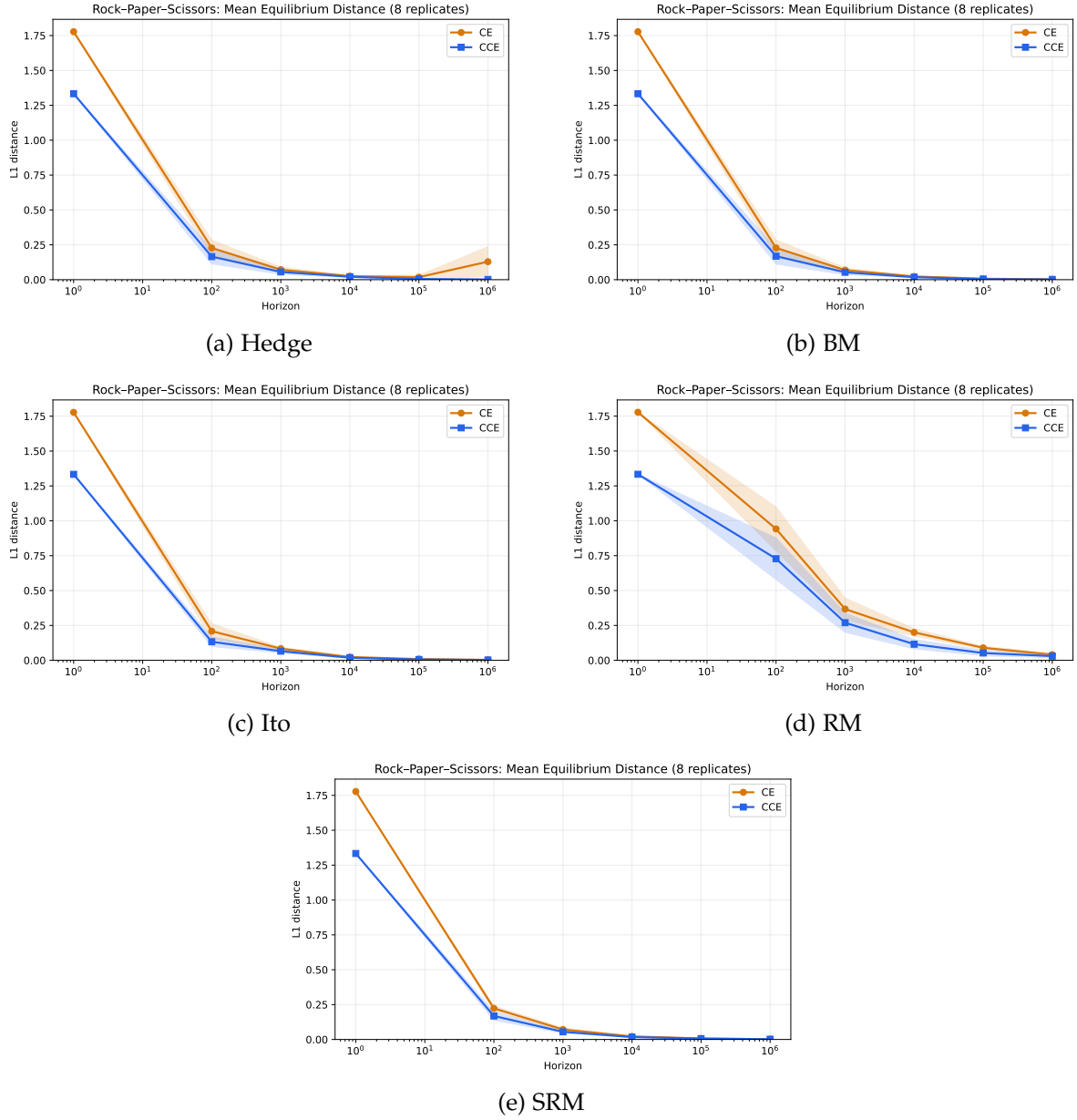
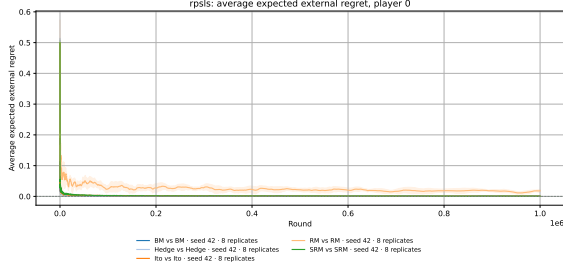
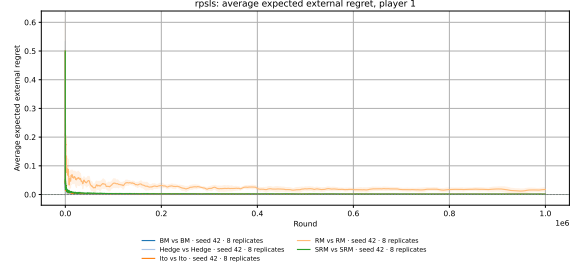


Figure 5.7.: RPS self-play, full information: mean L_1 distance of the empirical joint-action distribution to the CE and CCE sets. Panel order is Hedge, BM, Ito, RM, and SRM.

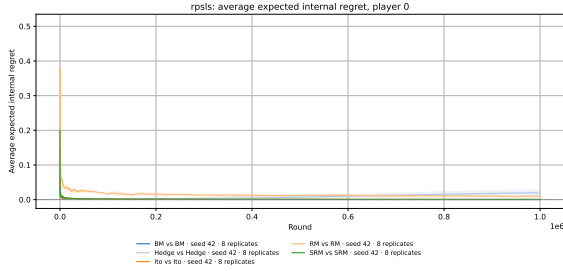
5. Experimental Results



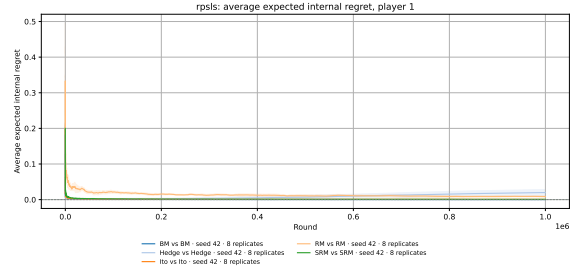
(a) Player 0: average external



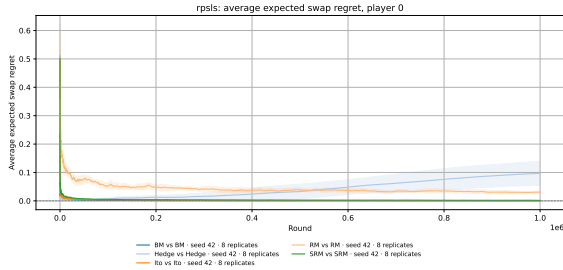
(b) Player 1: average external



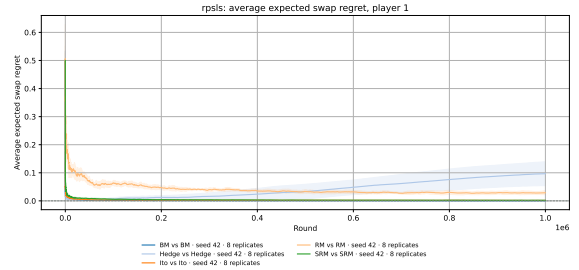
(c) Player 0: average internal



(d) Player 1: average internal



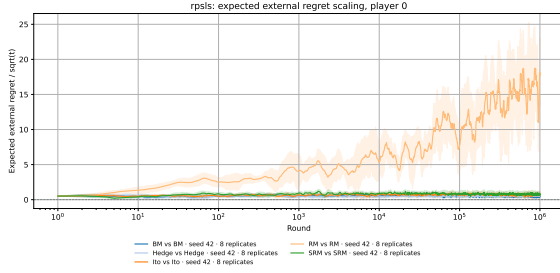
(e) Player 0: average swap



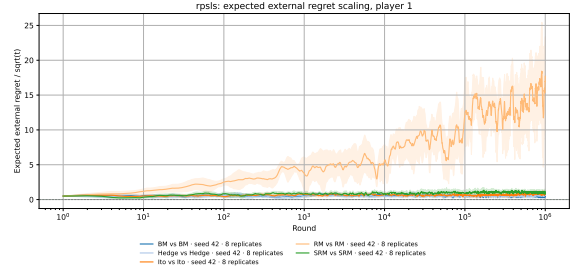
(f) Player 1: average swap

Figure 5.8.: RPSLS self-play, full information, $T = 10^6$, eight replicates: all average strategy-weighted regret results.

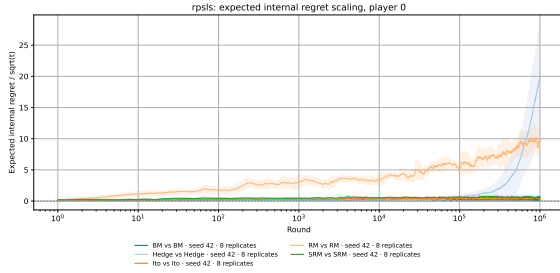
5. Experimental Results



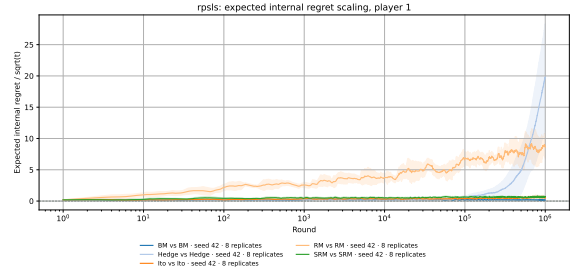
(a) Player 0: external / $\sqrt{t}$



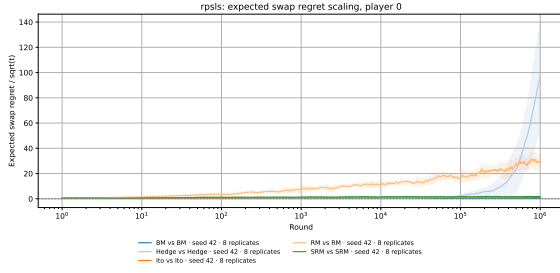
(b) Player 1: external / $\sqrt{t}$



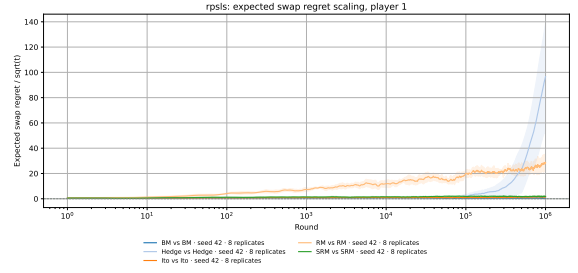
(c) Player 0: internal / $\sqrt{t}$



(d) Player 1: internal / $\sqrt{t}$



(e) Player 0: swap / $\sqrt{t}$



(f) Player 1: swap / $\sqrt{t}$

Figure 5.9.: RPSLS self-play, full information: all normalized strategy-weighted regret results.

5. Experimental Results

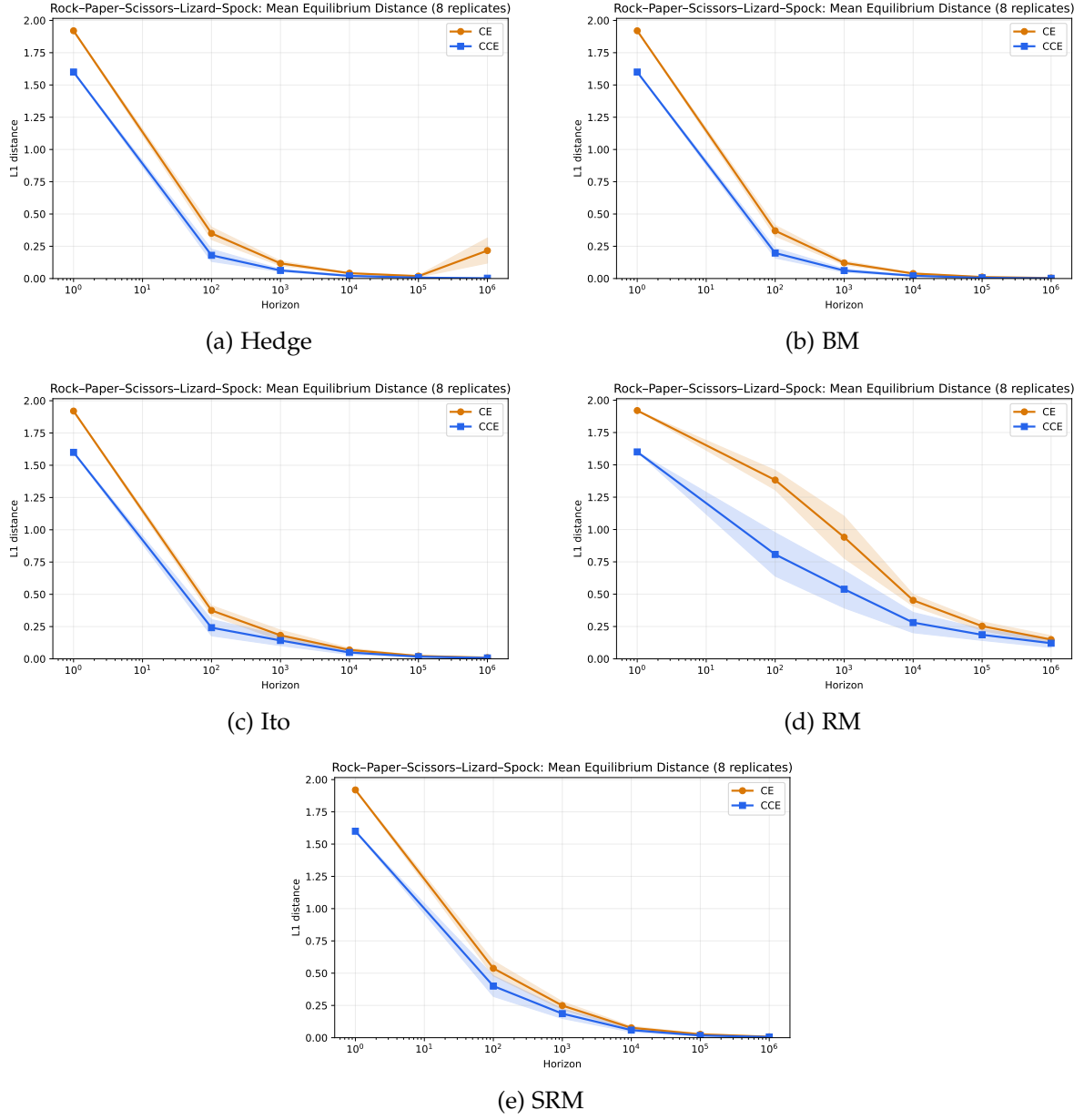


Figure 5.10.: RPSLS self-play, full information: mean L_1 distance to CE and CCE. Panel order is Hedge, BM, Ito, RM, and SRM.

5. Experimental Results

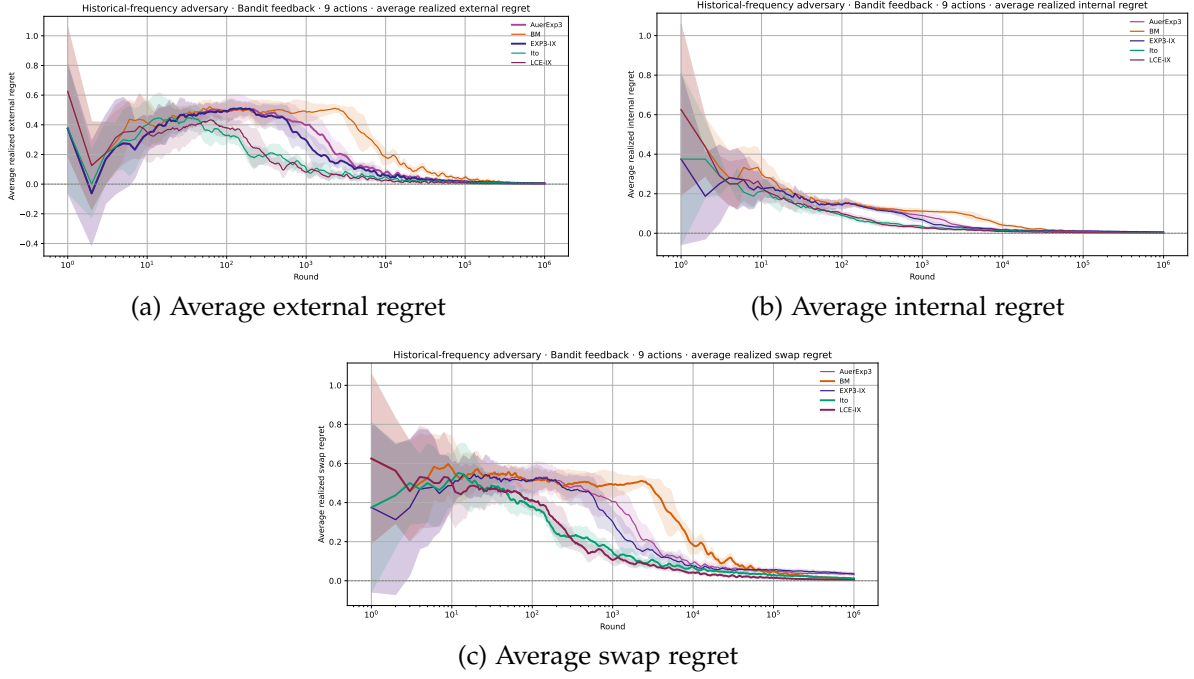


Figure 5.11.: Historical-frequency adversary, bandit feedback, $K = 9$, $T = 10^6$, eight replicates: complete average realized regret results.

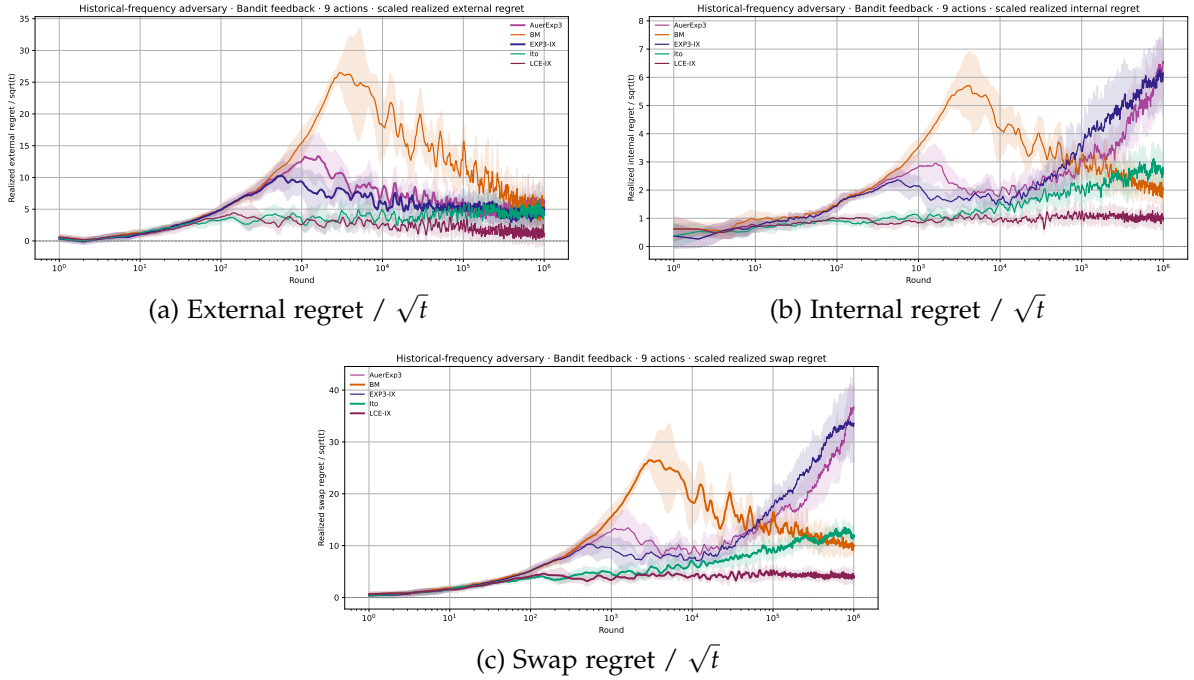


Figure 5.12.: Historical-frequency adversary, bandit feedback: complete normalized realized-regret results.

regret methods no longer have a large uniform advantage. One plausible explanation is the persistence of the reward process: each action moves by at most one grid step per round, so a currently good action tends to remain competitive locally, while the reward dynamics are independent of the learner’s actions. This explanation is a hypothesis rather than a measured property of the present experiment; switching rates or reward autocorrelations would be needed to test it directly.

RPS self-play. The RPS bandit results are shown completely in Figures 5.15–5.17. The regret panels separate the two external-regret baselines from BM, Ito, and LCE-IX more clearly on internal and swap regret than on external regret. The equilibrium panels show the corresponding game-theoretic behavior. The external-regret algorithms are guaranteed only toward CCE, whereas the swap-regret constructions are associated with CE convergence. In this particular game some external-regret runs also become close to CE, which is possible because the guarantee is one-sided: external regret ensures CCE convergence but does not exclude better behavior in a specific game.

No stable ordering between BM and Ito is visible across all bandit environments. In RPS one can be faster over one part of the horizon and the other elsewhere. LCE-IX is the most consistently strong of the three swap-regret methods in the current bandit results.

RPSLS self-play. RPSLS produces the same broad external-versus-swap distinction with a larger action set (Figures 5.18–5.20). AuerExp3 and Exp3-IX have higher internal and swap regret over much of the run, while BM, Ito, and LCE-IX approach lower levels. The equilibrium-distance panels provide the same comparison in terms of empirical play. As in RPS, the exact BM–Ito ordering changes with the environment and with the horizon, so the present data do not justify ranking one reduction uniformly above the other. LCE-IX again appears among the fastest and most stable methods across the bandit regret and equilibrium views.

5.3. Cross-Experiment Observations

Several patterns recur across the complete set of figures.

Under full information, Hedge behaves as expected from an external-regret algorithm: its external-regret curves are substantially better than its internal and swap curves. In self-play, this distinction also appears in equilibrium space, where Hedge can become close to CCE while remaining farther from CE. BM and Ito are useful controls for this comparison because they use external-regret learners internally but change the master update through a swap-regret reduction. Their swap-regret and CE-convergence behavior is correspondingly much stronger than standalone Hedge in the present experiments.

RM and SRM should not be interpreted as two implementations of the same theorem. The original RM result of Hart and Mas-Colell [3] is a joint self-play CE-convergence statement. The stationary procedure gives an individual pairwise-regret guarantee. The experiments reproduce this difference qualitatively: RM can perform poorly in the single-agent adaptive

5. Experimental Results

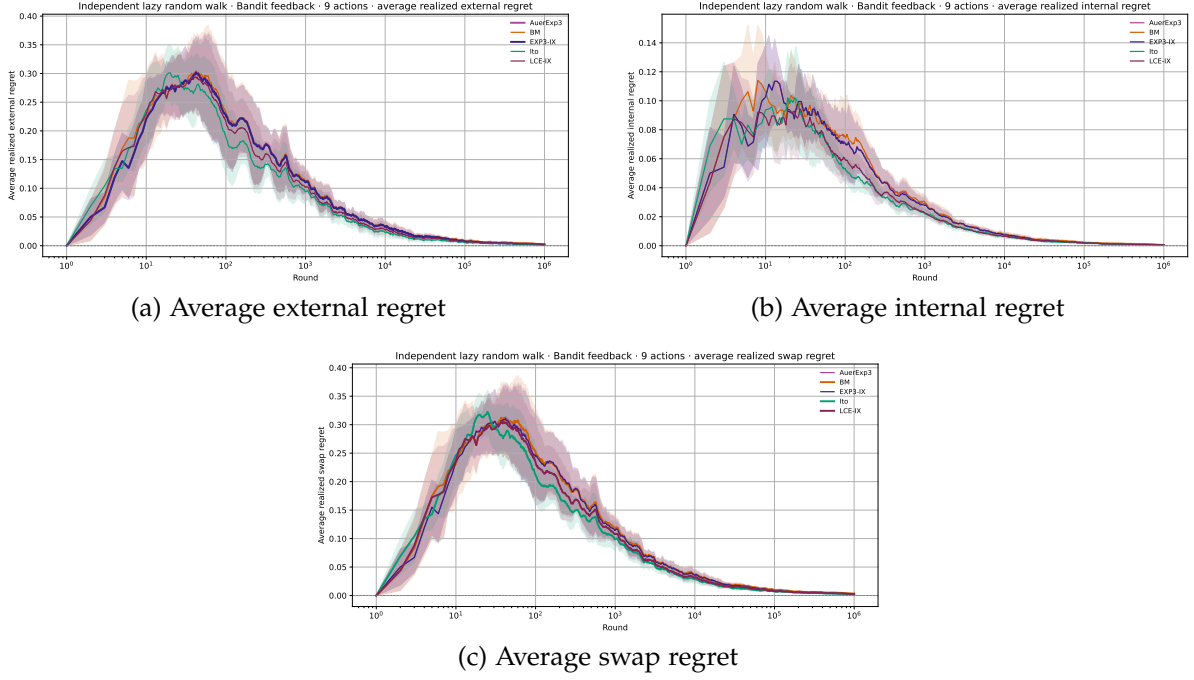


Figure 5.13.: Independent lazy random walk, bandit feedback, $K = 9$, $T = 10^6$, eight replicates: complete average realized-regret results.

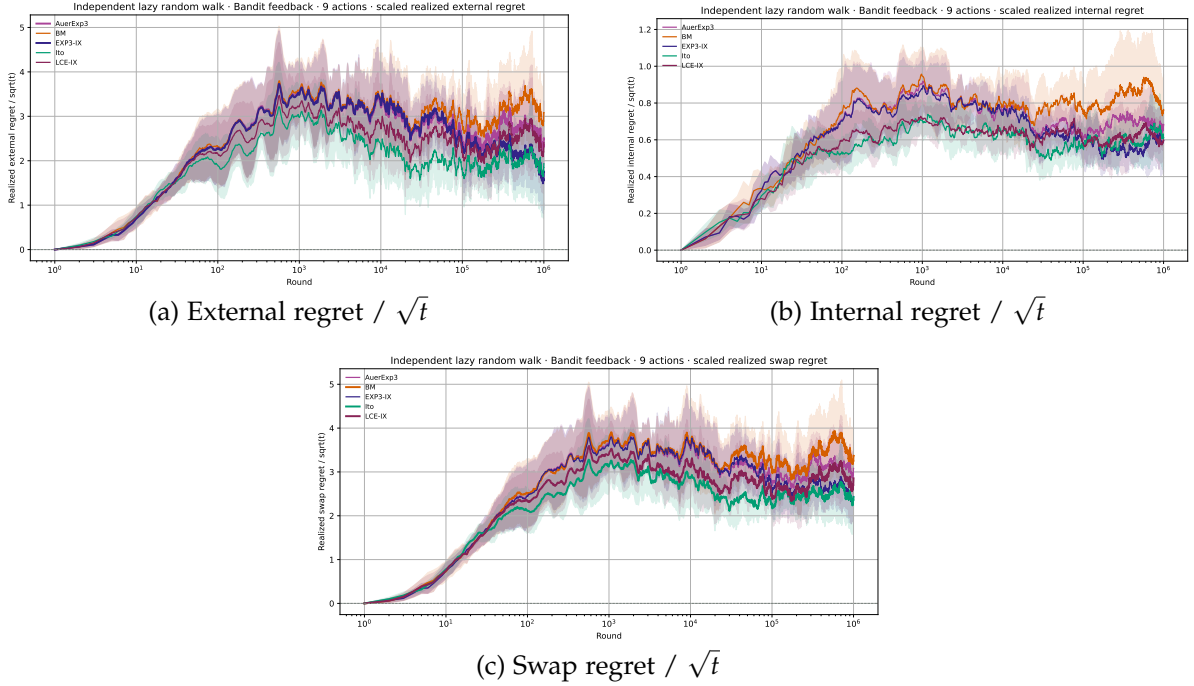
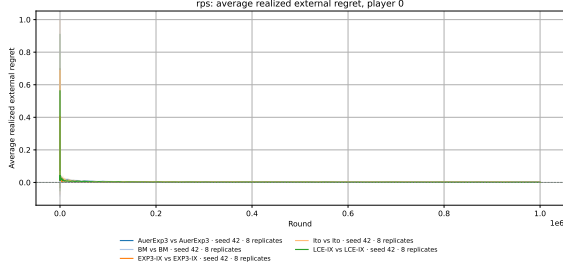
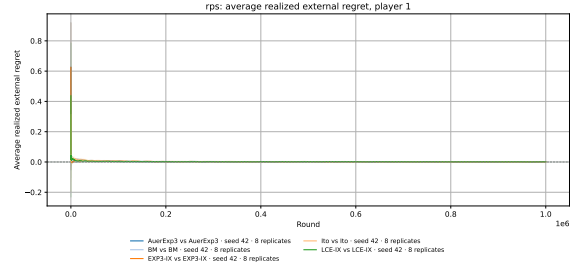


Figure 5.14.: Independent lazy random walk, bandit feedback: complete normalized realized-regret results.

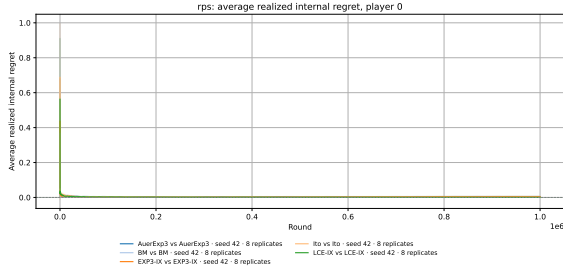
5. Experimental Results



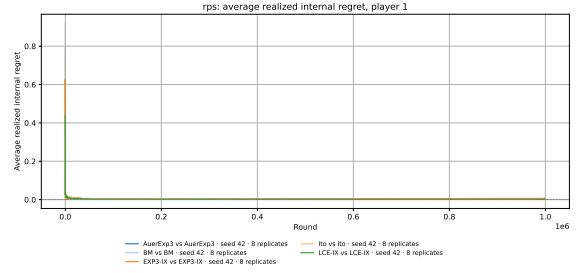
(a) Player 0: average external



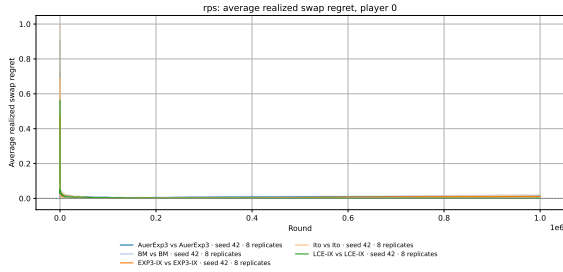
(b) Player 1: average external



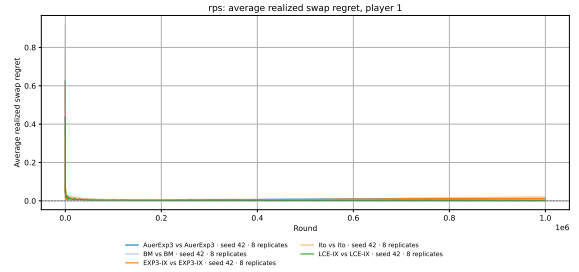
(c) Player 0: average internal



(d) Player 1: average internal



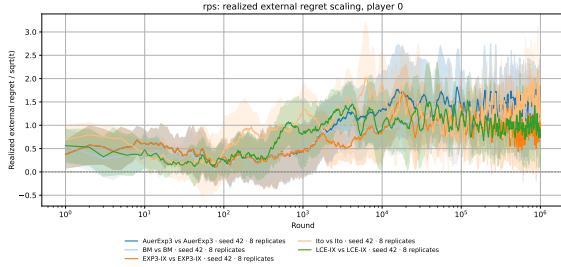
(e) Player 0: average swap



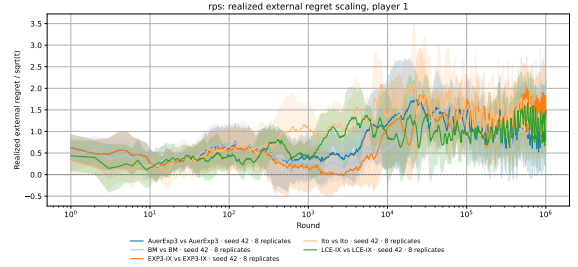
(f) Player 1: average swap

Figure 5.15.: RPS self-play, bandit feedback, $T = 10^6$, eight replicates: all average realized-regret results.

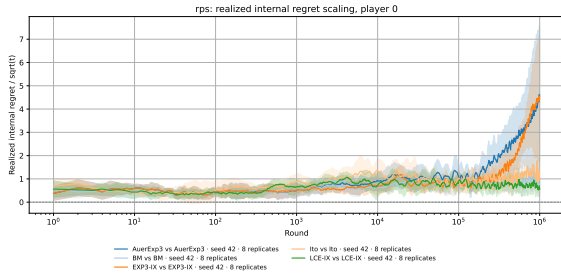
5. Experimental Results



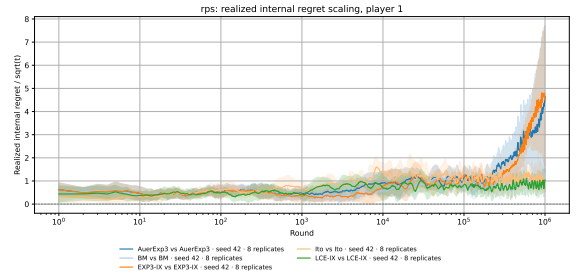
(a) Player 0: external / $\sqrt{t}$



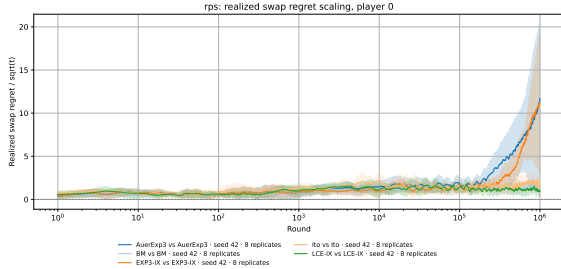
(b) Player 1: external / $\sqrt{t}$



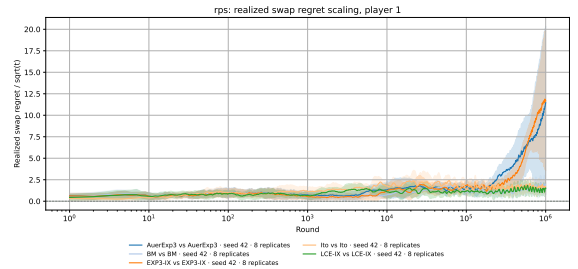
(c) Player 0: internal / $\sqrt{t}$



(d) Player 1: internal / $\sqrt{t}$



(e) Player 0: swap / $\sqrt{t}$



(f) Player 1: swap / $\sqrt{t}$

Figure 5.16.: RPS self-play, bandit feedback: all normalized realized-regret results.

5. Experimental Results

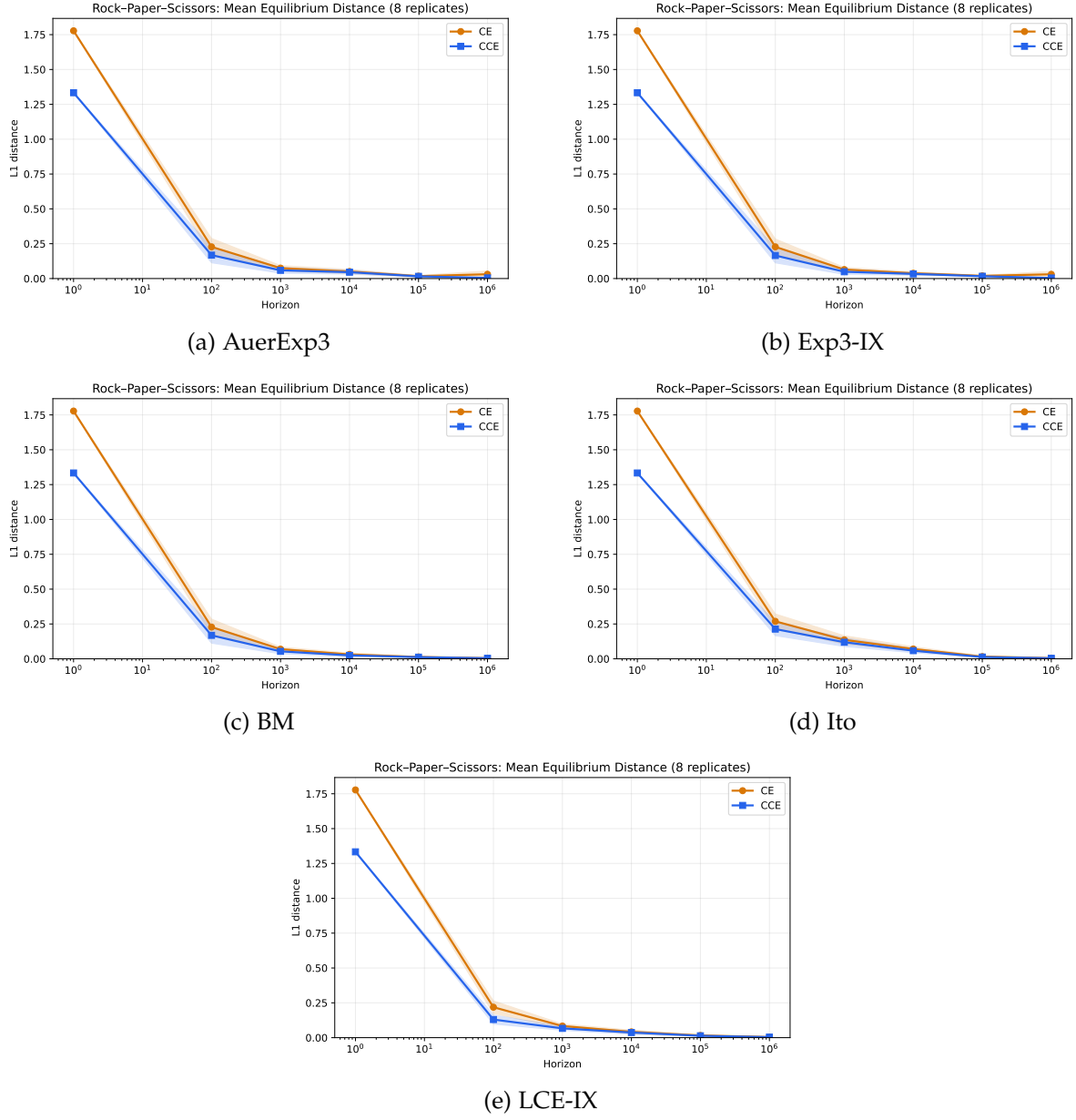
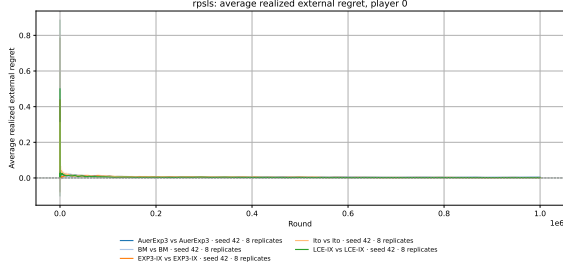
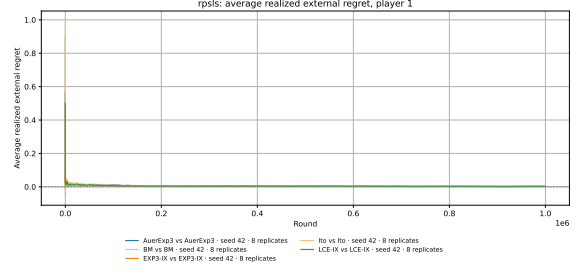


Figure 5.17.: RPS self-play, bandit feedback: mean L_1 distance to CE and CCE. Panel order is AuerExp3, Exp3-IX, BM, Ito, and LCE-IX.

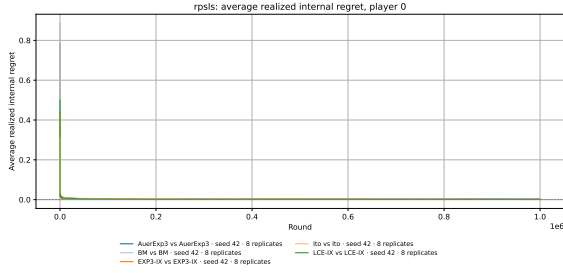
5. Experimental Results



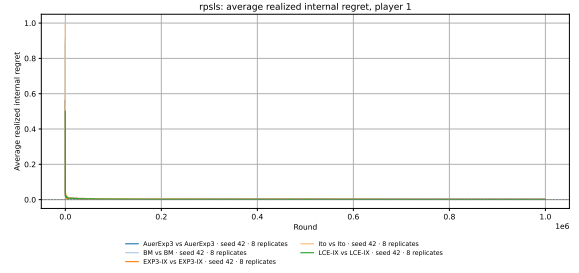
(a) Player 0: average external



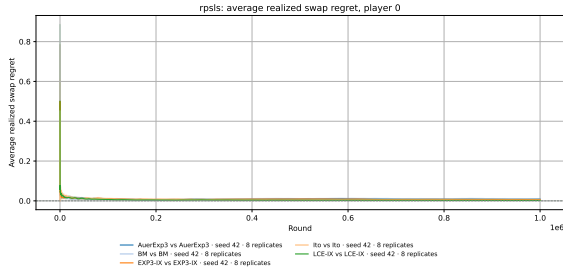
(b) Player 1: average external



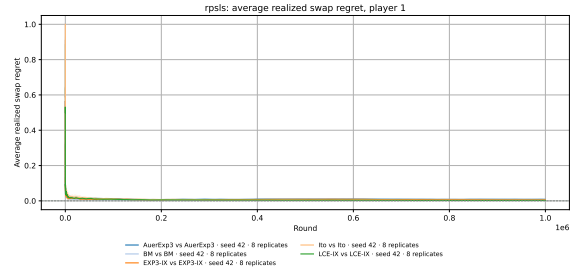
(c) Player 0: average internal



(d) Player 1: average internal



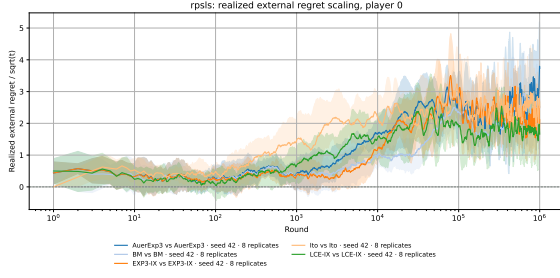
(e) Player 0: average swap



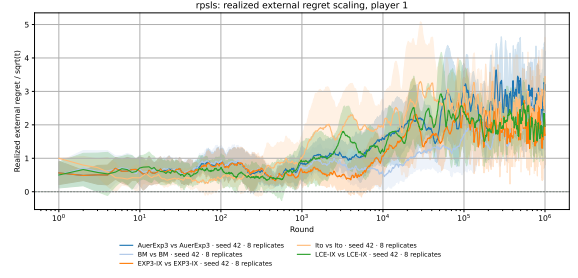
(f) Player 1: average swap

Figure 5.18.: RPSLS self-play, bandit feedback, $T = 10^6$, eight replicates: all average realized-regret results.

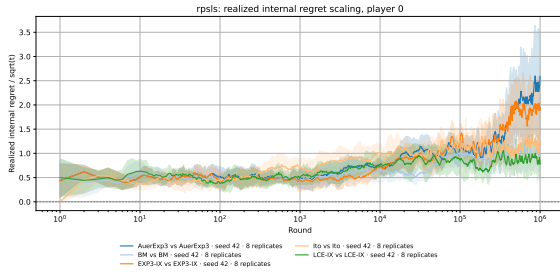
5. Experimental Results



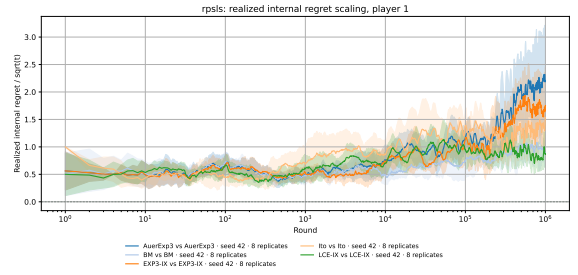
(a) Player 0: external / $\sqrt{t}$



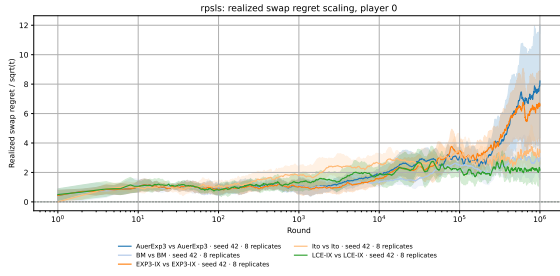
(b) Player 1: external / $\sqrt{t}$



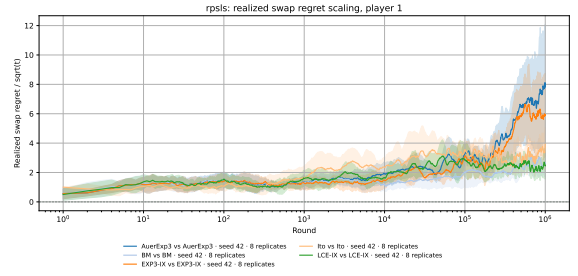
(c) Player 0: internal / $\sqrt{t}$



(d) Player 1: internal / $\sqrt{t}$



(e) Player 0: swap / $\sqrt{t}$



(f) Player 1: swap / $\sqrt{t}$

Figure 5.19.: RPSLS self-play, bandit feedback: all normalized realized-regret results.

5. Experimental Results

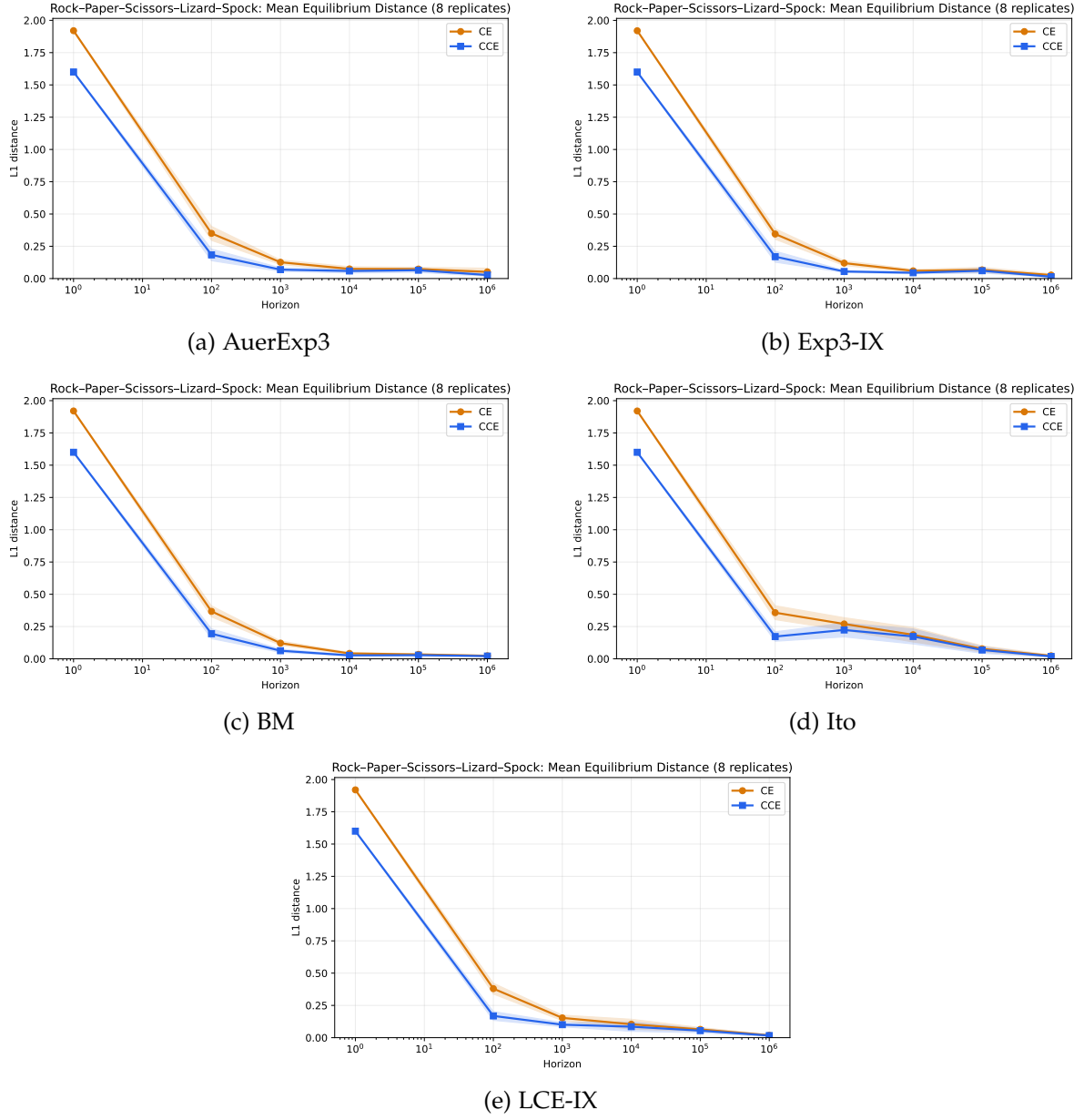


Figure 5.20.: RPSLS self-play, bandit feedback: mean L_1 distance to CE and CCE. Panel order is AuerExp3, Exp3-IX, BM, Ito, and LCE-IX.

regret experiments while RM self-play still approaches CE; SRM is consistently strong in both the one-player and self-play settings. Because the action sets are finite, SRM’s vanishing internal regret also implies vanishing swap regret through $R_T^{\text{swap}} \leq KR_T^{\text{int}}$.

Under bandit feedback, the historical-frequency adversary and the self-play games expose a clear finite-horizon gap between the external-regret baselines and the three algorithms designed to control swap regret. The lazy random walk is the main exception: all methods behave similarly and even the external-regret baselines have small observed swap regret. The temporal persistence of the random walk is a plausible explanation, but the current experiment does not isolate that mechanism.

Finally, the data do not establish a uniform ordering between BM and Ito under bandit feedback. Their relative behavior changes across environments. LCE-IX is the most consistently strong bandit method in the current archive, usually combining low normalized swap regret with rapid equilibrium convergence. More replicates and additional environments are needed before turning these observations into a general ranking.

6. Discussion

6.1. Full-Information Results

The full-information experiments separate three different roles that are easy to conflate if only one regret measure is inspected. Hedge has the cleanest external-regret behavior, but its internal and swap regret are much larger. In the RPS and RPSLS self-play experiments, the same distinction appears in the empirical distribution: Hedge approaches the CCE set more closely than the CE set. This is consistent with the standard implication from external regret to coarse correlated equilibrium [2].

BM and Ito show why the reduction step matters. Both use external-regret learners internally, yet the master construction changes the relevant deviation class. Their observed swap regret is much smaller than standalone Hedge, and their self-play trajectories approach CE. In the current full-information runs, Ito usually reaches a low-regret regime earlier than BM. This is a finite-horizon observation rather than a theorem comparing the two reductions, but it is consistent across most of the present full-information environments.

The RM–SRM comparison requires a different interpretation. The original regret-matching theorem of Hart and Mas-Colell [3] states that if all players use RM, the empirical distribution of play converges almost surely to CE. It is not an individual adversarial no-internal-regret theorem. The stationary procedure, by contrast, gives vanishing pairwise regret for an individual player without requiring the opponents to use the same rule. The experiments make this difference visible. RM can have relatively poor regret curves in the one-player adaptive environments and still move toward CE in self-play when both players use RM. SRM is strong in both settings. Since the action set is finite, its internal-regret guarantee also implies vanishing average swap regret through the relation $R_T^{\text{swap}} \leq KR_T^{\text{int}}$.

One should therefore avoid reading the RM self-play result as evidence for a standalone RM external-regret guarantee. What the figures show is narrower and more interesting: substantial positive cumulative regret at finite horizons can coexist with empirical movement toward CE in joint RM self-play. The relevant Hart–Mas-Colell theorem concerns the empirical distribution generated when all players follow the procedure.

6.2. Bandit Results and Environment Dependence

The historical-frequency adversary gives the clearest bandit separation. AuerExp3 and Exp3-IX are designed for external regret, and their realized swap regret is much larger than that of BM, Ito, and LCE-IX over the tested horizon. The three latter algorithms do not share exactly the same theoretical guarantee—Blum–Mansour is usually stated in terms of swap

pseudo-regret, Ito in expectation, and LCE-IX has a stronger high-probability statement—but all are constructed to control swap deviations [4, 13, 14]. The experiment does not prove that the external-regret baselines fail no-swap-regret asymptotically; it shows that the missing guarantee matters in this finite-horizon adaptive environment.

The independent lazy random walk gives almost the opposite picture. All five bandit algorithms lie much closer together, including on the swap-regret plots. A plausible explanation is the temporal persistence built into the environment: each action reward changes by at most one grid step in a round, and the reward process is independent of the learner’s action. A currently good action is therefore likely to remain competitive for several nearby rounds, and the environment does not react to predictable learner behavior. The current results support this interpretation only indirectly. A follow-up experiment could measure best-action switching frequencies or reward autocorrelation to test it more directly.

The two self-play games again separate the external-regret baselines from the swap-regret methods, although less uniformly than the historical-frequency adversary. The regret curves and equilibrium-distance plots both show that the stronger deviation control can translate into faster movement toward CE. At the same time, an external-regret algorithm may also become close to CE in a particular game; the external-regret guarantee only promises CCE convergence and does not exclude stronger behavior on an easier instance.

No consistent ordering between BM and Ito emerges under bandit feedback. Their relative finite-horizon performance changes between HFA, RPS, and RPSLS. This is one place where additional environments would be more informative than a stronger claim from the present data. LCE-IX is more stable across the current bandit experiments: it is usually among the lowest-regret methods and its equilibrium distance decreases quickly in both self-play games.

6.3. Limitations and Further Experiments

The main quantitative limitation is the number of replicates. Eight runs are sufficient for the large separations visible in several figures, but some confidence intervals remain wide and the current data are not suited to fine ranking when means are close. The principal comparisons should therefore be rerun with more replicates before the final version if time permits.

The experiment set is also deliberately small. RPS and RPSLS provide transparent cyclic self-play examples, but they do not cover general-sum games. The historical-frequency adversary and the lazy random walk give two qualitatively different one-player environments, but they do not span the range of possible adaptive or stochastic reward processes. More environments would be particularly useful for the BM–Ito comparison and for testing whether the strong LCE-IX behavior persists.

The normalized $R_t/\sqrt{t}$ plots should not be interpreted as a formal empirical rate experiment. Several algorithms use parameters chosen for the final horizon $T = 10^6$, so a prefix at time t is not equivalent to an independently tuned run whose total horizon is t . A controlled scaling study would require separate experiments at several final horizons.

Finally, the figures in the current archive were generated immediately before the implementation-version-3 cleanup. The cleanup was intended to restore literature-faithful numerical details rather than to change the algorithms, but the final thesis should regenerate the main configurations with the frozen implementation version so that the code and reported results match exactly.

7. Conclusion

The experiments show that the distinction between external, internal, and swap regret is visible in practice, but the strength of the separation depends strongly on the environment. Under full information, Hedge consistently has the most favorable external-regret profile while its internal and swap regret are much larger. BM and Ito demonstrate that external-regret learners can be combined through a reduction to obtain markedly better swap-regret behavior. SRM is also consistently strong in the present finite-action experiments.

The RM results require a different reading. Ordinary regret matching is not used here as a standalone adversarial no-internal-regret algorithm. Its classical guarantee is that when all players use the procedure, empirical play converges to correlated equilibrium [3]. The self-play experiments are consistent with this result: RM approaches CE even though its finite-horizon regret curves can be much less favorable than those of SRM. The stationary variant has the stronger individual pairwise-regret guarantee and performs well in both the one-player and self-play experiments.

Under bandit feedback, the historical-frequency adversary and the self-play games produce the clearest gap between AuerExp3/Exp3-IX and the dedicated swap-regret methods. The independent lazy random walk largely removes this gap, showing that stronger worst-case guarantees do not imply a large advantage on every reward sequence. BM and Ito have environment-dependent finite-horizon behavior, while LCE-IX is the most consistently strong bandit method in the current archive.

The equilibrium-distance figures give the game-theoretic counterpart of the regret comparison. Hedge self-play can become close to CCE while remaining farther from CE, whereas the internal- and swap-oriented procedures generally approach CE more directly. The RM experiments further show why the scope of a theorem matters: joint self-play CE convergence and an individual adversarial regret guarantee are different statements.

The present results should be treated as a complete preliminary experiment set rather than as a final ranking of the algorithms. The next steps are to rerun the main configurations with the frozen implementation version, increase the replicate count for the central comparisons, and add further environments where the current data leave BM and Ito difficult to order. Those additions would strengthen the quantitative conclusions without changing the basic comparison developed in this thesis.

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A. Additional Experimental Material

The project archive contains the complete set of regret and equilibrium-distance figures generated for the current experiment round. The main text uses only the plots needed for the comparisons discussed in Chapter 5. Additional figures can be added here in the final version if further replicates, horizon-scaling experiments, or new games materially change the conclusions.